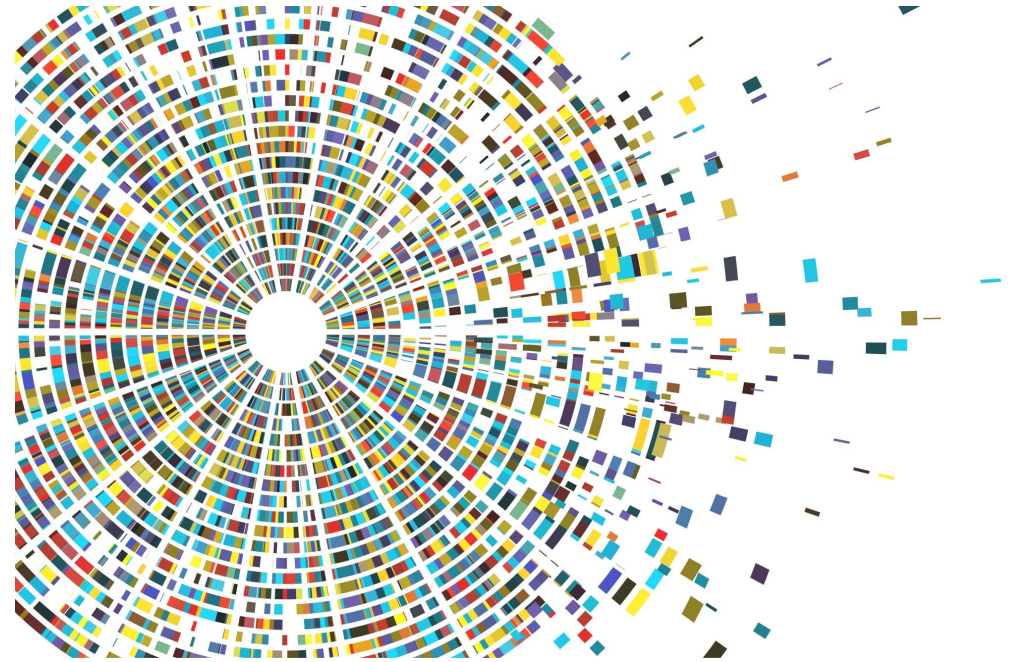


Use Case 8 · 300.AI

Virtual Account Receivables Exception Resolution Agent

A human-supervised agentic MVP for receivables exception investigation

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IBM

CONTEXT

Business problem

What virtual accounts do

Virtual accounts provide customer- or purpose-specific identifiers linked to a physical bank account, enabling granular sub-ledgering and easier receivables reconciliation. They support centralised cash management and automated reconciliation.

The manual cost today

Analysts examine payment records, virtual account mappings, invoices, customer details and operating procedures manually before deciding where a payment belongs — delaying cash credit, extending unallocated-cash ageing and producing inconsistent case documentation.

Why a payment lands in the exception queue

- 1 The virtual account is inactive, expired or incorrectly configured
- 2 The virtual-account-to-customer mapping is missing
- 3 The amount differs from the expected invoice amount
- 4 The payer name or reference is inconsistent
- 5 Several invoices could match the same payment
- 6 A duplicate or previously allocated payment is suspected

SCOPE

Use-case definition

Primary user	Virtual account or receivables operations analyst
Trigger	An inbound payment cannot be automatically allocated
Objective	Investigate the exception and recommend the most likely customer and invoice allocation
Agent actions	Plan the investigation, retrieve evidence, validate the virtual account, find candidate invoices, rank possible matches, apply the approved procedure and prepare a case summary
Output	Recommended allocation, confidence score, supporting evidence, exception reason and draft operations note
Human decision	Approve, revise, reject or escalate the recommendation
Permitted write	Create or update a case-management task after approval



Prohibited actions

- Autonomous ledger posting
- Account credit
- Payment return
- Virtual account creation or modification



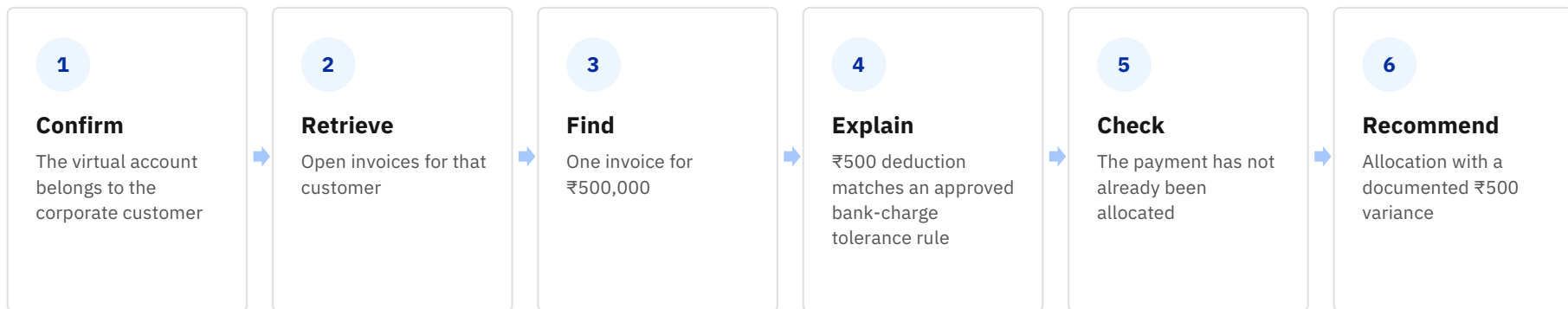
The analyst decides

Every recommendation is a proposal. Nothing reaches the ledger without a human decision.

HOW IT BEHAVES

Worked example

A corporate customer expects ₹500,000 into a dedicated virtual account. **A payment of ₹499,500 arrives with an incomplete reference and is routed to the exception queue.**



The agent sends the recommendation to an analyst for approval. It does not post the money or change the invoice.

PROPOSED ACTORS

Agent roles



Triage Agent

Classifies the exception and creates an investigation plan



Virtual Account Agent

Validates the account's status, ownership and effective dates



Matching Agent

Retrieves candidate customers and invoices, applies approved matching rules



Policy Agent

Finds the applicable exception-handling procedure and tolerance rules



Validation Agent

Checks evidence, duplicate risk, contradictions and confidence



Case Agent

Produces the analyst summary and draft case note



Operations Analyst – human in the loop

Receives the Case Agent output and makes the final allocation decision, or asks the agents for further evidence



Deterministic rules engine – not an agent

Amount variance, tolerance evaluation, duplicate detection and date checks run in testable code, never in agent reasoning

DESIGN PRINCIPLE

Why this is agentic — and where it stops



Adaptive — the agent reasons

More than a fixed matching workflow. The agent can:

- Create an investigation plan based on the exception type
- Select the required systems and retrieval tools
- Adapt its search when the initial customer or invoice match fails
- Compare competing allocation hypotheses
- Retrieve the relevant operating procedure
- Detect missing or contradictory evidence
- Ask for analyst input when confidence is insufficient



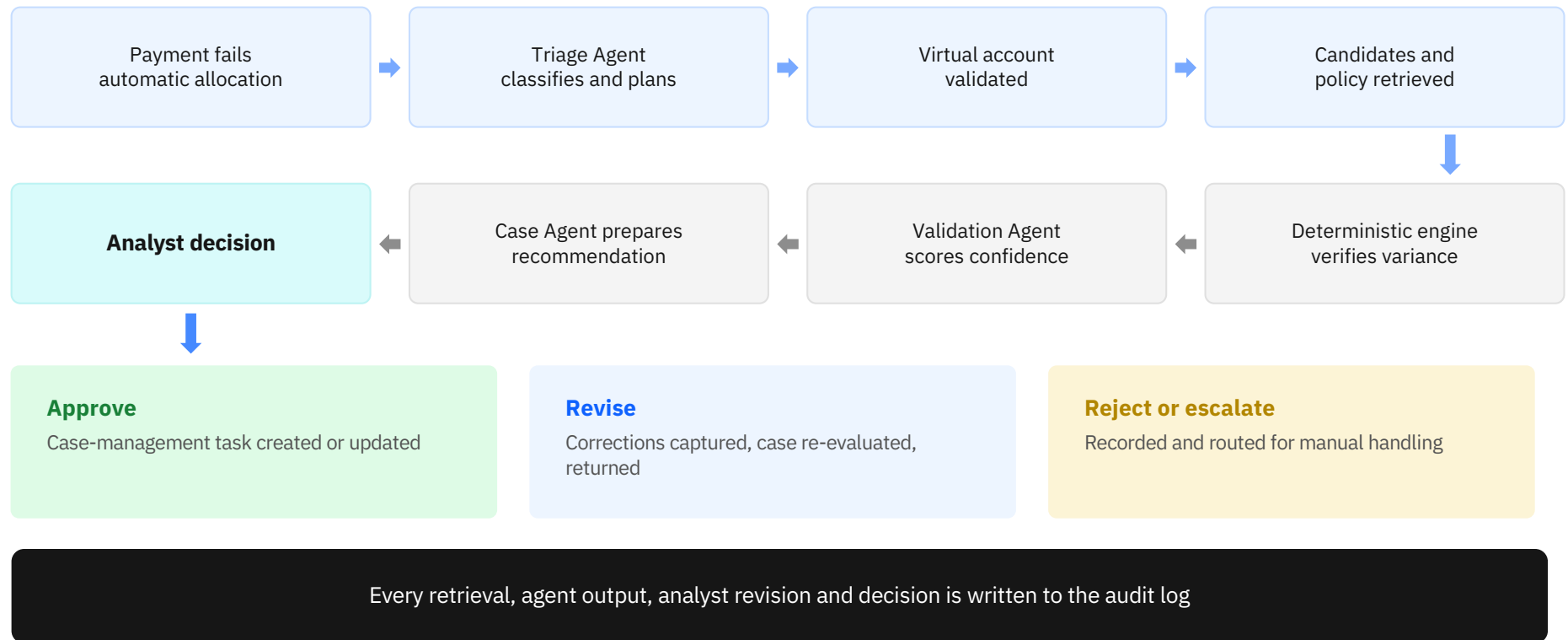
Deterministic — code and humans decide

Final financial actions never depend on model judgement:

- Variance arithmetic and tolerance evaluation run in testable code
- Duplicate and prior-allocation checks are deterministic
- A candidate that fails verification cannot be shown as a recommendation
- Low-confidence and conflicting cases are always escalated
- Analyst approval is required before any write action
- Payer name and remittance text are treated as data, never as instruction

HOW IT RUNS

Process flow



EDGE CASES

Exception and escalation behaviour



Missing or invalid mapping

Account status recorded with evidence. Identification from payer name or reference is marked as inference, not mapping, and the case is escalated as low-confidence.



Amount or reference mismatch

The rules engine computes the variance and compares it with the approved tolerance. An unexplained variance is never absorbed — it is escalated with the figure shown.



Suspected duplicate payment

Prior allocation status is confirmed deterministically. The agent recommends nothing and presents both payment records side by side.



Multiple competing candidates

All verified candidates are presented with their evidence. The agent does not choose between them on unsupported grounds.



Insufficient evidence or low confidence

An explicit "no recommendation" outcome is returned with the evidence gathered and the reason. The agent does not invent an allocation to produce an answer.



Analyst requests revision

The correction and its rationale are captured, the case is re-evaluated, and the revised recommendation returns for decision.

DELIVERY

One-month MVP scope

Boundaries

- One currency
- One corporate customer
- Three exception categories: invalid mapping, amount or reference mismatch, suspected duplicate
- One approved exception-handling procedure and tolerance table

Read-only sources

- Payment transaction repository
- Virtual account master
- Customer master
- Open invoice repository

Build

- Deterministic matching and tolerance rules
- Agent reasoning over the four sources
- Analyst approval interface: approve, revise, reject, escalate
- Case creation in a test case-management system
- Audit log of retrievals, outputs and decisions

Out of scope for the MVP: autonomous posting or account credit · payment return or repair · virtual account changes · master-data changes · collections action · sanctions and financial crime screening · multi-currency and multi-customer operation · production deployment

GOVERNANCE

Controls and safeguards



Decision boundary

- Analyst approval is mandatory before any case-management write action
- All source-system integrations are read-only for the MVP
- Amount, variance, tolerance and duplicate checks are deterministic and independently testable
- Every recommendation is traceable to source evidence and an explicit procedure or tolerance rule
- The agent must not fabricate customer identifiers, invoice numbers or amounts



Evidence, data and access

- Low-confidence, conflicting and duplicate-suspected cases are always escalated
- Payer name, remittance reference and narrative are externally controlled text — treated as data, never as instruction, and surfaced to the analyst if they attempt to direct the agent
- Access controls and data masking applied at source level
- Tool access restricted to minimum required permissions
- Agent tool calls, outputs, analyst revisions and decisions are auditable

Read-only for the MVP. The only write action the solution performs is creating a case-management task after an analyst has approved it.

EVALUATION

Proposed MVP acceptance criteria

Measure	Target
Exception classification accuracy	At least 90% on the agreed test set
Top-candidate accuracy	At least 85% for supported exception types
Allocation precision	At least 95% of recommendations presented as confident are correct
Incorrect-allocation rate (<i>Optional</i>)	Reported explicitly — a wrong allocation is materially worse than an escalation
Correct escalation	Low-confidence, conflicting and duplicate cases escalated in at least 90% of applicable cases
Evidence coverage	100% of recommendations include source evidence
Variance verification	100% of recommended allocations carry a verifiable variance calculation
Financial safety	No autonomous financial posting or virtual account change
Auditability	Every data retrieval, recommendation and analyst decision is auditable

Precision and the incorrect-allocation rate are the decisive measures: a wrong allocation mis-states a customer ledger, while an escalation costs an analyst a few minutes.

RECOMMENDED PLAN

Four-week delivery plan

Week 1

Define and verify

- Fix currency, customer, exception categories and sources
- Agree procedure, tolerance table and confidence thresholds
- Build the deterministic rules engine and its tests
- Assemble the labelled test set

Week 2

Retrieve and match

- Read-only retrieval tools
- Triage and investigation planning
- Virtual account validation
- Candidate matching and structured outputs

Week 3

Judge and oversee

- Policy retrieval and confidence scoring
- Escalation paths
- Analyst approval interface and case creation
- Audit trail, access control, failure handling

Week 4

Evaluate and demo

- Run the full evaluation set
- Correct rules and prompts, re-run
- User acceptance testing
- MVP demonstration

The rules engine is built in Week 1, before the agents — the checker exists before the thing being checked

HONEST CAVEATS

Assumptions and limitations

One-month target applies to a controlled prototype

Existing APIs, mock connectors or de-identified data — not live payment systems

Production deployment is a later phase

Live integration and operational sign-off follow the MVP

Accuracy is bounded by scope

Assessed only for three exception categories, one currency and one corporate customer

Recommendations are proposals

Agent output is never an autonomous financial action

Learning from corrections is deferred

The MVP captures corrections in a controlled feedback dataset but does not adapt behaviour from them — an unevaluated learning loop inside a financial control is itself a control risk